Analysing University Library Space Utilisation and Student Study Patterns

Oluwadurotimi Alade

University of Ibadan

September 2026

Presentation Overview

1. Introduction
2. Research Questions
3. Literature Review
4. Data and Methodology
5. Preliminary Results
6. Methodology
7. Results and Discussion

Background

- University libraries provide spaces for individual and collaborative study.
- The way students use these spaces can vary across zones, time periods, and days.
- Understanding these patterns can support better use of available study spaces.

Problem Statement

Library study spaces have limited seating capacity, but demand may vary across different zones, time periods, and days. A structured analysis of occupancy, study duration, student entries, and satisfaction can help identify patterns in space utilisation.

Aim and Objectives

Aim

To analyse university library space utilisation and student study patterns using synthetic observational data.

Objectives

1. Measure occupancy rates across library study zones.
2. Compare utilisation across time periods and days.
3. Examine study duration, student entries, and satisfaction.
4. Explore relationships among selected study-pattern variables.
5. Develop practical recommendations from the observed patterns.

Research Questions

1. How does library space occupancy vary across the three study zones?
2. How does occupancy vary across different time periods?
3. How does occupancy vary across days of the week?
4. What patterns exist between student entries, study duration, and occupancy?
5. What relationship exists between occupancy and student satisfaction?

Literature Review

- Physical library space can influence how students choose and use study areas (Cha and Kim 2015).
- Students use academic library spaces for different individual and collaborative study activities (Bedwell and Banks 2013).
- Students' needs and preferences can vary across different library spaces and study environments (May and Swabey 2015).

Conceptual Background

- Library space utilisation reflects how available study spaces are occupied and used by students.
- Students may select different study environments depending on their study activities and preferences (Cha and Kim 2015).
- Academic library spaces can support both individual and collaborative study activities (Bedwell and Banks 2013).
- The design and characteristics of study spaces can influence students' experience of the library environment (DeFrain and Hong 2020; Hegde, Boucher, and Lavelle 2018).

Dataset Overview

Dataset Feature	Value
Observation days	6
Observation periods per day	4
Study zones	3
Total observations	72

- The dataset covers a hypothetical Monday-to-Saturday observation week.
- Each observation represents one study zone, one period, and one day.
- The dataset is fully synthetic and embedded in the project.

Variables and Data Structure

Variable	Description
Day	Day of observation
Period	Two-hour observation period
Zone	Library study zone
Capacity	Number of available seats
Occupied Seats	Number of occupied seats during observation
Avg. Study Duration	Average study duration in minutes
Student Entries	Number of student entries recorded
Satisfaction Score	Student satisfaction rating on a 1–5 scale

- Occupancy rate is derived from Occupied Seats and Capacity.
- The raw observations are embedded in `data.tex`.

Study Design

- Data type: fully synthetic benchmark data.
- Observation period: one hypothetical week, Monday to Saturday.
- Study zones: Quiet, General, and Group.
- Four two-hour observation periods were used each day.
- Total observations: 72.

Variables

Variable	Description
Day	Day of observation
Period	Two-hour observation period
Zone	Library study zone
Capacity	Available seats in the zone
Occupied Seats	Seats occupied during observation
Study Duration	Average study duration in minutes
Student Entries	Student entries recorded during the period
Satisfaction	Average satisfaction score from 1 to 5

Observation Periods

Period	Time
P1	8:00–10:00 AM
P2	10:00 AM–12:00 PM
P3	2:00–4:00 PM
P4	4:00–6:00 PM

Each observation covers one study zone during one two-hour period on one day.

Preliminary Occupancy Result

The current validated dataset gives an overall mean occupancy rate of:

$$\boxed{71.59\%}$$

Research Design and Methodology

- The study uses a descriptive and comparative observational design.
- The observation period covers a hypothetical Monday-to-Saturday week.
- Observations were recorded across three study zones and four two-hour periods each day.
- Each observation represents one study zone, one period, and one day.
- The analysis focuses on occupancy, study duration, student entries, and satisfaction.
- All observations are synthetic and embedded in the project for benchmark reproducibility.

Statistical Analysis Approach

- Occupancy rate is calculated as:

$$\text{Occupancy Rate} = \frac{\text{Occupied Seats}}{\text{Capacity}} \times 100$$

- Descriptive statistics are used to summarise occupancy, study duration, student entries, and satisfaction.
- Mean occupancy is compared across study zones, observation periods, and days.
- Pearson correlation is used to examine linear associations between the quantitative variables.
- Results are interpreted as descriptive associations rather than causal effects.

Descriptive Analysis

Measure	Overall Mean
Occupancy Rate	71.59%
Average Study Duration	76.53 minutes
Student Entries	22.60
Satisfaction Score	3.14 / 5

- Mean occupancy across the 72 observations was 71.59%.
- Average study duration was 76.53 minutes.
- The mean number of student entries was 22.60.
- Mean satisfaction was 3.14 on the 1–5 scale.

Comparative and Correlation Analysis

Mean Occupancy by Zone

Zone	Occupancy
Quiet	79.69%
General	73.54%
Group	61.53%

Mean Occupancy by Period

Period	Occupancy
P1	62.36%
P2	74.26%
P3	66.34%
P4	83.38%

Pearson Correlations

Variable Pair	r
Occupancy–Duration	0.063
Occupancy–Entries	0.508
Occupancy–Satisfaction	-0.013
Duration–Entries	0.104
Duration–Satisfaction	0.726
Entries–Satisfaction	0.018

- P4 had the highest mean occupancy.
- Occupancy and student entries showed a moderate positive association.
- Study duration and satisfaction showed a strong positive association.

Results and Discussion

- The Quiet Zone recorded the highest mean occupancy at 79.69%, followed by the General Zone at 73.54% and the Group Zone at 61.53%.
- P4 recorded the highest mean occupancy at 83.38%, indicating greater space use during the final observation period.
- Saturday recorded the lowest mean daily occupancy at 64.65%, while Wednesday recorded the highest at 74.93%.
- Occupancy and student entries showed a moderate positive association ($r = 0.508$).
- Study duration and satisfaction showed a strong positive association ($r = 0.726$).
- These findings describe patterns and associations within the synthetic benchmark dataset and do not establish causal relationships.

Recommendations and Limitations

Recommendations

- Review space allocation in the Quiet Zone because it recorded the highest mean occupancy.
- Pay particular attention to demand during P4, which recorded the highest mean occupancy.
- Consider study duration when evaluating student experience, given its positive association with satisfaction.
- Continue monitoring occupancy and student use across individual zones and periods.

Limitations

- The dataset is fully synthetic and represents one hypothetical observation week.
- The analysis covers three study zones and four observation periods per day.
- The observational design identifies patterns and associations but does not establish causation.

Conclusion

- The analysis shows clear differences in library space utilisation across study zones and observation periods.
- The Quiet Zone recorded the highest mean occupancy, while the Group Zone recorded the lowest.
- P4 was the busiest observation period, while Saturday had the lowest mean daily occupancy.
- Occupancy was moderately associated with student entries, while study duration showed a strong positive association with satisfaction.
- The findings provide a basis for targeted monitoring and planning of library study spaces.

Thank you

Questions?